



tery packs. If you make any of these battery packs, they're still considered first-life packs.

- Not many refiners are available to extract the metals like cobalt, lithium, and graphite, which are not naturally available in India to be mined.

- On the international front, there are restrictions being imposed by the countries where these are available.

- Ensuring that standards are met for collection, and ensuring safe handling and transportation is another challenge.

The opportunities

Here is the list of opportunities that could give rise to new businesses:

- There's a huge potential for material recycling, which is nascent and will be increasing multi-fold in future.

- Batteries will not just die like that after use in electric vehicles. They will be first used in their second life before going for recycling, because these have high-end, high-purity, and high-material cells.

- Direct recycling, that is, cathode to cathode recycling, is possible and it is most favourable for LFP. So that is a technology on which the recyclers need to work in India jointly.

- India has a huge opportunity for battery-grade refiners, the technology for which is yet to come to India from the countries like Japan, China, and Korea.

- There is a significant market for second-life batteries because 60-80% of energy is left in the cells, which can be used in toys, inverters, etc. This presents a big opportunity for dealers, service centres, and other authorised channels.

- Foreign investors are showing a lot of interest in the recycling sector in India due to India's huge consumer and increasing consumption of lithium-ion batteries.

- Indian recycling firms have the chance to partner with foreign

companies and get exposed to technology as well as entire network of partners.

Government's role

Here's the list of things expected from the Central and state governments, as these are all policy-related:

- Skilling the workforce and explaining them the chemistry of lithium-ion is important.

- Lithium-ion battery dismantling and recycling is a different game altogether and requires different regulations in place. Removing lithium is not the same as removing PCB from a mobile phone.

- There needs to be a mechanism to incentivise and ensure that the recycling is done properly and that the customers can return their batteries and collect the money.

- Battery Waste Management Rules, 2022 are a step in the right direction for the recyclers, refurbishers, producers, and importers. They prohibit landfill disposal and incineration of depleted batteries. The efficacy of implementation is going to be the most important factor, however.

- The centralised online portal is crucial to fix end-to-end responsibility. Making sure that there is traceability of the batteries through this online portal is going to be critical.

- There should be some kind of check for not just the kind of extraction and efficiency of the technology, but also for the emissions.

- We have a lot of things to learn from the lead-acid battery market. Lead in batteries had to be handled carefully, but its implementation become very localised, dominated by the recyclers.

- It's better to make and announce policy beforehand to prevent germination of the unorganised sector in this domain.

- Government should help standardise the procedure and set a benchmark.

Way forward

Here are some suggestions for the industry:

- Look at the waste as a raw material.

- We don't manufacture many lithium-ion batteries. So, whatever gets recycled, rather than exporting to foreign refiners, we should refine it within our country and benefit from that.

- We need to have a strong reverse logistic system to develop a supply-chain network and let the aggregators and partners work as collectors.

- The recycling cost will be high, but the benefits will be higher, but for that the scale has to be achieved, and one needs to have over 90% efficiency extraction targets.

- Batteries from electric 2-wheelers and electric 3-wheelers can be used for stationary applications like solar storage stations, or can be directly sold to manufacturers in the toy industry.

- If they are in good shape and are not damaged, 4-wheeler battery cells can be used for electric 2-wheeler and 3-wheeler batteries, which can run well for 18-21 months.

- The recyclers should form an association to fix uniform pricing and to procure the raw materials. The customers, who are the end users, should get a reasonable buyback value so that it becomes an entirely sustainable ecosystem.

- International entities can set up their recycling facilities in India with the help of local partners, which could help introduce the latest techniques here.

- Recyclers should work together instead of competing with each other. That will help in regulating battery prices and increase profitability for the recyclers. **EFY**

Vaishali Yadav, a journalist at EFY, muses upon clouds, security, sustainability, integration, smartness, machines and learning—both trivially and technologically.